Collapse-Retained Information and Representation-Independent Pushdown Exposure Complexity

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Abstract

Higher-order semantic freedom need not disappear when all lower-order information has been fixed. We ask what physical cost is forced when such retained information is subsequently erased by one deterministic semantic action. The motivating examples come from universal k-fibers of optimal compatibility in periodic shortest-path systems, where one common exact distance and one common compatibility truncation through order $k-1$ coexist with all $2^{\binom{s}{k}}$ order-k patterns on a distinguished s-simplex.

We introduce *collapse-retained information* and a physical resource, *source-stack exposure depth*, measuring how many cells of the initial semantic representation must become visible before a common canonical target is reached. For an arbitrary finite semantic collapse family of size N, let D sources fail to hit the chosen canonical target, let X_h canonical sources be forced to expose deeper than h, and let $V(h)$ be the number of physical source observations available at depth h. Under a natural hidden-tail noncoalescence condition we prove the filtered capacity inequality

$$D + X_h + V(h) \geq N.$$

For unrestricted deterministic ε-pushdown realizations with arbitrary history-dependent encodings and arbitrary finite replacement words, hidden-tail noncoalescence is automatic. If q_{src} source control states and g_{src} source stack symbols occur on the collapse interface, then

$$D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq N.$$

The optimal universal lower envelope of the ordered exposure spectrum is determined: if $e_{(1)} \leq \cdots \leq e_{(K)}$ are the canonical source depths, then

$$e_{(r)} \geq \min\Big\{ h : q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq r \Big\},$$

and this rank-by-rank bound is simultaneously attained by a zero-push cleanup family. Thus the exposure law is pointwise sharp, not merely asymptotically sharp.

We then prove a self-contained universal-retention theorem. In a full periodic k-fiber, every lower-order structural prefix is an affine translation and hence injective. Fixing one k-face to the nongeodesic branch therefore leaves exactly

$$2^{\binom{s}{k}-1}$$

distinct sources that collapse under one final action. Consequently every exact pushdown realization obeys a representation-independent state/alphabet/exposure/debt tradeoff with retained information $\binom{s}{k} - 1$ bits. We derive a strong converse and direct-sum consequences: whenever depth-h physical capacity is deficient by λ bits, at least a $1 - 2^{-\lambda}$ fraction of the collapse family must pay debt or deeper exposure; under a growing deficit, asymptotically almost every universal witness is hard on approximately half of all k-face queries, equivalently on almost all of its absent-face queries.

Keywords. pushdown automata; physical complexity; source-stack exposure; representation debt; universal k-fibers; geodesic compatibility; information erasure; strong converse.

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1 Introduction

A deterministic pushdown representation can hide semantic information in many places: finite control, stack symbols, stack depth, and history-dependent encodings. Consequently, a lower bound tied to one canonical representation does not explain whether the observed cost is intrinsic. The present paper studies a more representation-independent question. Suppose a semantic system contains a large family of distinct sources that one action sends to one common target. How much of the physical source representation must be locally available, exposed, or left as representation debt before an exact pushdown realization can perform that collapse?

The question arose through a longer line of work beginning with a completion-aware 2:1 deterministic pushdown automaton. In that model the residual

$$\rho(w) = N_1(w) - 2N_0(w)$$

turns shortest completion into a directed arithmetic problem. Generalizing the residual to

$$\rho_{p,q}(w) = qN_1(w) - pN_0(w)$$

led in one direction to unrestricted, representation-independent state/stack lower bounds. In another direction, shortest-completion geometry led to periodic translation systems, higher-order geodesic compatibility, and universal k-fibers in which all lower-order optimality data are frozen while the next-order layer varies freely. The present paper reconnects those two branches.

Figure 1 records the concrete 2:1 origin. It is included only as research genealogy. No transition rule, state organization, stack encoding, completion convention, or bounded-replacement property of that automaton is assumed anywhere below.

The key distinction is between *static information capacity* and *collapse-retained information*. A family may contain many semantic variants, yet a lower-order prefix could identify most of them before the physical question becomes relevant. What matters for a collapse lower bound is the information still carried by distinct semantic sources immediately before one common action erases their distinction. Universal periodic k-fibers have an especially strong retention property: lower-order structural prefixes are affine translations, so they preserve all witness distinctions.

The second key distinction is between *stack height* and *source-stack exposure*. Stack-height complexity measures how large the current store becomes during a computation. Our quantity asks a different question: how many cells of the *initial semantic representation* must become physically visible before canonicalization? An implementation may push arbitrarily many administrative cells without increasing source exposure; conversely, a computation can have zero push complexity while exposing an arbitrarily deep initial stack.

Contributions. The paper establishes the following theorem package.

- We formulate a filtered collapse-capacity principle

$$D + X_h + V(h) \geq N,$$

  comparing semantic collapse load with depth-h physical observation growth and representation debt.

- For unrestricted deterministic ε-pushdown realizations we prove hidden-tail noncoalescence and obtain

$$D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq N.$$

  No realtime, bounded-replacement, canonical stack encoding, or history-independent representative is assumed.

- We determine the complete ordered exposure spectrum and prove pointwise sharpness for generic finite collapse systems.

- We give a self-contained universal k-fiber witness and prove exact face-collapse retention

$$N = 2^{\binom{s}{k}-1}.$$

- We derive a strong converse in bits and simultaneous-query/direct-sum bounds, including a typical-case statement for universal fibers.

The rest of the paper is organized so that the new physical theorem is logically independent of the universal construction. Sections 2–4 develop finite semantic collapse and pushdown exposure. Sections 5–6 construct and apply universal k-fibers. Section 7 gives strong-converse and direct-sum consequences, and Section 8 positions the resource against nearby notions.

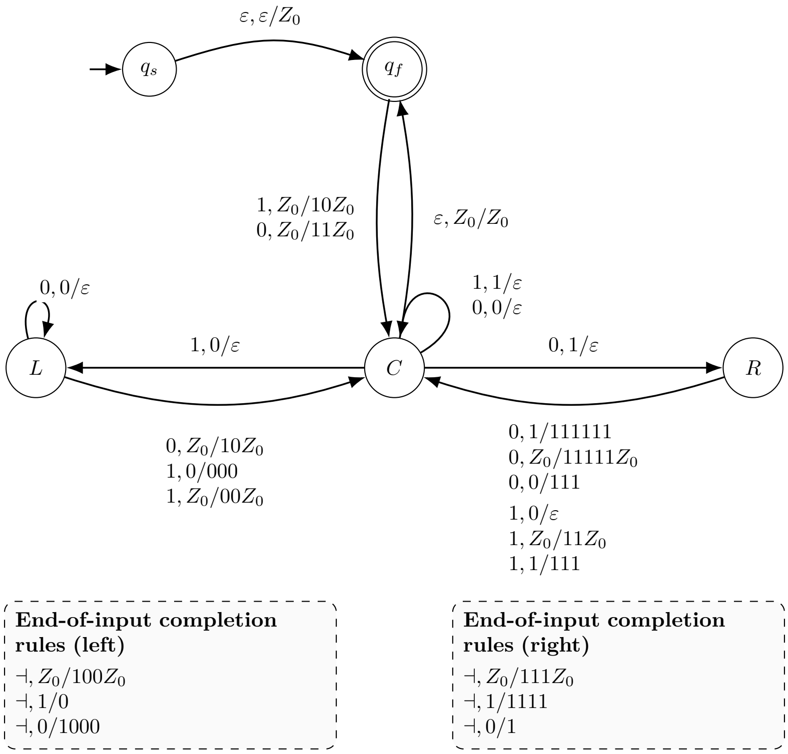

Figure 1: The completion-aware 2:1 DPDA that supplied the concrete discovery origin. The figure is motivational only; none of its physical representation choices enters the theorems below.

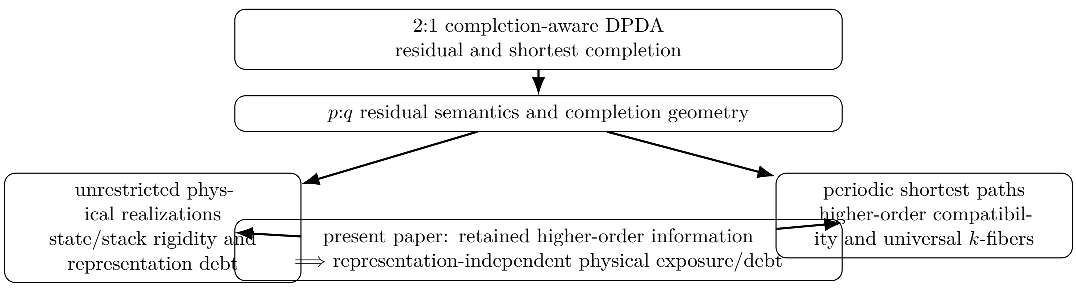

Figure 2: The two research branches reunited in the present paper.

2 Finite semantic collapse and filtered physical capacity

2.1 Semantic collapse families

Let M be a set of semantic states and A a finite action alphabet. For each $a \in A$, let

$$U_a : M \rightharpoonup M$$

be a deterministic partial transition.

Definition 2.1 (Collapse family)**.** *Fix $a \in A$ and $m^* \in M$. A nonempty finite set*

$$\mathcal{S} \subseteq U_a^{-1}(m^*)$$

*is an a-*collapse family *with target m^*. Its* collapse load *is $N = |\mathcal{S}|$, and its retained information is*

$$I_{\mathrm{ret}}(\mathcal{S}) = \log_2 N.$$

The definition intentionally refers to the source family immediately before the common transition. It therefore measures retained, continuation-relevant information rather than the cardinality of an earlier ambient fiber.

2.2 Exact physical lifts and canonical hits

Let P be a physical configuration space equipped with a semantic projection $\pi : P \to M$ and a section $\tau : M \to P$ satisfying $\pi \circ \tau = \mathrm{id}_M$. A physical configuration is called *settled* when no internal move is enabled. A *deterministic exact lift* may use internal moves and input-consuming moves, subject to the following requirements on the interface considered here: internal moves preserve π; from every relevant source representative the forced one-symbol computation is finite, consumes exactly one semantic action a, consumes no later input, and reaches a settled configuration in the fiber $\pi^{-1}(U_a(m))$. Thus exactness concerns semantic behavior, not return to a preferred physical encoding. The section selects one reachable reference representative from each relevant semantic fiber; other histories may realize the same semantic state by different physical configurations. For a chosen collapse target m^* we additionally choose $\tau(m^*)$ to be settled.

Fix an a-collapse family $\mathcal{S}$ and target m^*. Starting from $\tau(x)$ with unread input a, follow the unique forced raw computation through any internal moves, consume exactly a, and continue through forced internal moves until the next settled boundary, without consuming any later input. The termination clause above makes this macrostep well defined, and exactness guarantees that its endpoint lies in $\pi^{-1}(m^*)$.

Definition 2.2 (Canonical hit and debt)**.** *A source $x \in \mathcal{S}$ is a* canonical hit *if this one-symbol physical macrostep ends at the chosen representative $\tau(m^*)$. Otherwise its endpoint still represents m^* by exactness, but is noncanonical; then x is a* debt source*. Write*

$$D = \#\{x \in \mathcal{S} : x \textit{ is a debt source}\}, \qquad K = N - D.$$

Debt is deliberately interface-local: it records the failure to close the semantic collapse at the selected canonical target. A later normalization may repay that debt, but it has not disappeared at the collapse boundary.

2.3 Observation filtrations

Let

$$\omega_h : \tau(\mathcal{S}) \longrightarrow \Omega_h, \qquad h = 0, 1, 2, \ldots,$$

be a nested family of physical source observations: ω_{h+1} refines ω_h, so there is a forgetful map $r_h : \Omega_{h+1} \to \Omega_h$ with $\omega_h = r_h \circ \omega_{h+1}$. Write

$$V(h) = |\Omega_h|$$

for the corresponding observation-growth function. A canonical source computation is called *h-confined* when it never accesses physical source information beyond the part represented by ω_h before its first canonical hit. Let

$$X_h = \#\{x \in \mathcal{S} : x \text{ is canonical but not } h\text{-confined}\}.$$

Definition 2.3 (Hidden-tail noncoalescence). *The filtration has* hidden-tail noncoalescence *on the chosen collapse interface if two distinct canonical sources with the same ω_h-observation cannot both remain h-confined and reach the same chosen physical target $\tau(m^*)$.*

This condition is the abstract content needed for the counting theorem. For top-access pushdown stores it follows from the fact that an untouched hidden suffix cannot be changed without first becoming visible; this nontrivial verification is given in Section 3.

Theorem 2.4 (Filtered Collapse-Capacity Theorem). *If hidden-tail noncoalescence holds, then for every $h \geq 0$,*

$$\boxed{D + X_h + V(h) \geq N.}$$

Equivalently,

$$X_h \geq \big(K - V(h)\big)_+.$$

Proof. Among the K canonical sources, exactly $K - X_h$ are h-confined. Hidden-tail noncoalescence makes ω_h injective on this subset, so

$$K - X_h \leq |\Omega_h| = V(h).$$

Substituting $K = N - D$ gives the claim. □

Corollary 2.5 (Rank form). *Assume zero debt and that every canonical computation is h-confined for some finite h. For each source let $e(x)$ be the least such h, and sort these depths as*

$$e_{(1)} \leq \cdots \leq e_{(N)}.$$

With generalized inverse

$$V^{-1}(r) = \min\{h \geq 0 : V(h) \geq r\},$$

one has

$$\boxed{e_{(r)} \geq V^{-1}(r)}$$

for every $1 \leq r \leq N$.

Corollary 2.6 (Bit strong converse). *Let*

$$I_{\text{ret}} = \log_2 N, \qquad H_{\text{phys}}(h) = \log_2 V(h).$$

Then

$$\boxed{\frac{D + X_h}{N} \geq \left(1 - 2^{H_{\text{phys}}(h) - I_{\text{ret}}}\right)_+.}$$

In particular, if $\lambda \geq 0$ and $H_{\text{phys}}(h) \leq I_{\text{ret}} - \lambda$, then

$$\frac{D + X_h}{N} \geq 1 - 2^{-\lambda}.$$

3 Unrestricted deterministic pushdown exposure

3.1 Raw model

We use the standard deterministic top-access ε-pushdown interface. A physical configuration is

$$(q, \alpha), \qquad q \in Q, \quad \alpha \in \Gamma^*,$$

with the leftmost symbol of α regarded as the top. A raw transition inspects the finite-control state, the visible top symbol when one exists, and either one input symbol or ε, and replaces the visible top symbol by an arbitrary fixed finite word. Determinism includes the usual exclusion that an enabled ε-move prevents a competing input-consuming move at the same state/top mode. No bound is imposed on replacement length. A permanent bottom marker may be used and is not counted among the work symbols; the same argument also covers the empty-stack convention. The lift may contain arbitrarily many history-dependent physical representatives of the same semantic state. The section τ merely selects one reachable reference representative for the collapse interface; uniqueness of representation is not assumed.

Only the initial source stack is charged to the exposure resource. Conceptually tag the work cells present at the source boundary, from top to bottom, by immutable provenance labels $1, 2, \ldots$. These labels are bookkeeping devices only and are not available to the machine. When a transition replaces a visible cell by a finite word, the removed source cell keeps no descendant provenance: every newly written cell is administrative, even if it carries the same stack symbol. Thus arbitrary replacement words cannot create new source cells for purposes of the resource measure.

Definition 3.1 (Source-stack exposure depth)**.** *For a canonical source x, define $e(x)$ to be the largest provenance index of an initial source-stack cell that becomes physically visible as the top cell during the one-symbol macrostep, up to and including the settled canonical hit. If no initial work cell becomes visible, put $e(x) = 0$.*

Thus $e(x)$ is not maximum stack height and not the number of pushes. It is a depth of *read access to the initial semantic representation.*

For the chosen canonical sources, let

$$q_{\mathrm{src}} = \bigl|\{\text{source control state of } \tau(x)\}\bigr|$$

and let g_{src} be the number of work-stack symbols that actually occur in their initial source stacks, with the convention $g_{\mathrm{src}} = 1$ if every canonical source stack is work-empty. If $K = 0$, set $q_{\mathrm{src}} = 0$ and $g_{\mathrm{src}} = 1$; then every exposure inequality below reduces immediately to $D = N$. Administrative states and symbols used only after leaving the source interface are not charged to these source-local parameters.

For $h \geq 0$ define the depth-h source signature as follows: record the source control state and, if the initial stack has fewer than h work cells, record it completely together with its end; otherwise record its top h source work symbols. The number of possible signatures is at most

$$C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^r.$$

Lemma 3.2 (Hidden suffix cannot coalesce)**.** *Two distinct canonical sources with the same depth-h source signature cannot both reach the same fixed canonical target while exposing at most h initial source cells.*

Proof. Suppose two source representatives share the same depth-h signature and both computations remain within the first h source cells. Until a deeper source cell becomes visible, every raw step sees the same control state, the same currently visible top symbol, and the same unread input status on both computations. Determinism therefore forces the same raw rule whenever the visible prefixes agree.

A raw rule replaces only the visible top symbol by a fixed finite word. Consequently it cannot alter the untouched source suffix below the exposed cut. Pushed administrative material may be created and later removed, but the hidden suffix remains literally unchanged until a deeper source cell is exposed. Hence two distinct hidden suffixes cannot be transformed into one identical complete target stack by computations that never expose those suffixes. If the hidden suffixes were already identical, equality of the complete source configurations together with section injectivity would identify the semantic sources. Both alternatives contradict distinctness. Therefore at most one shallow canonical source occupies each depth-h signature. □

Theorem 3.3 (Source-Stack Exposure Profile)**.** *For every finite semantic collapse family realized by an unrestricted deterministic ε-pushdown system and every $h \geq 0$,*

$$\boxed{D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r} \geq N.}$$

No bounded-replacement, realtime, homogeneous-tail, canonical residual, or unique-representation hypothesis is required.

Proof. Use the depth-h source signatures as the filtration in Theorem 2.4. Lemma 3.2 verifies hidden-tail noncoalescence, and the displayed signature count bounds $V(h)$. □

The global machine-resource version follows immediately from $q_{\mathrm{src}} \leq |Q|$ and $g_{\mathrm{src}} \leq |\Gamma|$.

4 Optimal exposure spectrum and pointwise sharpness

Let $K = N - D$ be the number of canonical sources and sort their exposure depths:

$$e_{(1)} \leq e_{(2)} \leq \cdots \leq e_{(K)}.$$

Define

$$d_r = \min \left\{ h \geq 0 : q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \geq r \right\}.$$

Theorem 4.1 (Exact exposure-rank lower bound)**.** *For every $1 \leq r \leq K$,*

$$\boxed{e_{(r)} \geq d_r.}$$

For $g_{\mathrm{src}} = g \geq 2$,

$$d_r = \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1,$$

whereas for $g_{\mathrm{src}} = 1$,

$$d_r = \left\lceil \frac{r}{q_{\mathrm{src}}} \right\rceil - 1.$$

Proof. At most C_h^{src} canonical sources can have exposure at most h by Theorem 3.3. If $e_{(r)} < d_r$, then with $h = e_{(r)}$ there are at least $r > C_h^{\mathrm{src}}$ shallow canonical sources, a contradiction. The closed forms invert the geometric sum. □

Corollary 4.2 (Aggregate exposure)**.**

$$\boxed{\sum_{x \text{ canonical}} e(x) \geq \sum_{r=1}^{K} d_r = \sum_{h \geq 0} \left(K - q_{\mathrm{src}} \sum_{j=0}^{h} g_{\mathrm{src}}^{j} \right)_{+}} .$$

For fixed $g \geq 2$ *and* $K/q_{\mathrm{src}} \to \infty$*,*

$$\bar{e} \geq \log_g (K/q_{\mathrm{src}}) - O_g(1),$$

while for $g = 1$ *one has* $\bar{e} = \Omega(K/q_{\mathrm{src}})$*.*

Theorem 4.3 (Pointwise sharpness)**.** *The rank lower bound is pointwise sharp. For every* $q \geq 1$*,* $g \geq 1$*, and* $H \geq 1$*, there exists a zero-debt deterministic* ε*-pushdown collapse family with*

$$K = q \sum_{r=0}^{H} g^r, \qquad q_{\mathrm{src}} = q, \qquad g_{\mathrm{src}} = g,$$

for which

$$\boxed{e_{(r)} = d_r \qquad (1 \leq r \leq K).}$$

Initial-segment truncations give every intermediate family size and remain rank-sharp with their actual source-local alphabet. The canonicalization uses no push operation.

Proof. Take source control states $q_1, \ldots, q_q$, one cleanup state c, a permanent bottom marker Z_0, and a work alphabet Γ of size g. Use every pair

$$(q_i, wZ_0), \qquad w \in \Gamma^*, \quad |w| \leq H,$$

as a distinct semantic source. One common semantic action a sends all sources to one target.

If $w \neq \varepsilon$, the physical a-transition removes the first source work symbol and enters c; deterministic ε-rules in c remove the remaining source word. If $w = \varepsilon$, the a-transition on Z_0 goes directly to the common canonical target. No rule pushes material. A source represented by a word of length r has exposure exactly r. There are exactly qg^r source representatives at depth r, hence exactly

$$q \sum_{j=0}^{h} g^j$$

with exposure at most h for $0 \leq h \leq H$. The ordered spectrum therefore equals the generalized inverse of this count at every rank. Truncating the list in nondecreasing word length preserves the same rank saturation; if the truncation has not yet used every work symbol, its actual g_{src} is smaller and the same counting argument applies to that actual source alphabet. □

Remark 4.4 (Why exposure is a distinct resource)**.** *The witness in Theorem 4.3 has zero push complexity during canonicalization but unbounded source exposure. Conversely, a machine may push arbitrarily much administrative material above the source without exposing any deeper source cell. Source exposure is therefore neither push count nor maximum current stack height.*

5 Periodic geodesic fibers and universal retained information

We now recall, in compact self-contained form, the universal semantic test family supplying an exponentially large collapse load while exact distance and all lower-order optimal-compatibility data remain fixed. Universal k-fiber freedom is a background platform here rather than a novelty claim of the present article; the new step is to identify what information survives to a common collapse boundary and convert it into physical exposure complexity.

5.1 Periodic action systems and geodesic compatibility

Let Φ be a finite phase set and I a finite action alphabet. Whenever action i is defined at phase ϕ, let

$$T_i(z, \phi) = (z + g_i(\phi), \tau_i(\phi)), \qquad g_i(\phi) \in \mathbb{Z}^n,$$

with positive integer cost $c_i(\phi)$. Fix a target state t. For a target-reachable state v, write $d(v)$ for minimum cost to t.

A finite action set J is a *structural cube* at phase ϕ when every ordering of every subset is executable, has the same endpoint and cost, and distinct subsets have distinct endpoints. For such a face, let $T_J v$ and c_J denote its common endpoint and cost. The face is *geodesic* at v when

$$c_J + d(T_J v) = d(v).$$

Write $H(v)$ for the resulting simplicial optimal-compatibility complex.

For fixed exact distance and fixed $H_{\le k-1}$, the possible order-k layers form the k-fiber. Call a periodic k-fiber *full universal on an s-simplex* if it contains labeled witnesses v_A, one for every $A \subseteq \binom{[s]}{k}$, with one common exact distance and one common complete truncation through order $k-1$, and with $H_{=k}(v_A)$ equal to the pattern A on the distinguished simplex. We first give an explicit witness, then use only this abstract full-universality property together with periodic prefix injectivity.

5.2 A universal witness construction

Fix $s \ge k \ge 2$, put

$$\mathcal{E} = \binom{[s]}{k}, \qquad m = |\mathcal{E}| = \binom{s}{k}, \qquad R = \binom{s-1}{k-1}.$$

Use two phases G, C and coordinates

$$\{b\} \sqcup \{p_1, \dots, p_s\} \sqcup \{h_E : E \in \mathcal{E}\}.$$

The target is $(0, C)$. For each $i \in [s]$, define a distinguished action $a_i : G \to G$ with translation

$$u_i = e_{p_i} + \sum_{E \ni i} e_{h_E}$$

and cost $R + 1 = \|u_i\|_1$. Add a switch $\sigma : G \to C$ of cost 1 and zero translation. In phase C, add unit-cost $\pm e_q$ completion actions for every coordinate q. No completion action is available in G, and no distinguished action is available in C. Consequently no structural face in phase G can mix the switch with a distinguished action: the ordering that applies the switch first would make the distinguished action unavailable.

For every $A \subseteq \mathcal{E}$, let $L = m + 1$ and define the witness state v_A in phase G by

$$(v_A)_b = -(L - |A|), \qquad (v_A)_{p_i} = -1,$$

$$(v_A)_{h_E} = -(k - 1 + \mathbf{1}_{E \in A}).$$

Lemma 5.1 (Metric and structural simplex)**.** *For x in phase C, $d(x, C) = \|x\|_1$, and for x in phase G,*

$$d(x, G) = 1 + \|x\|_1.$$

The distinguished actions span a structural s-simplex in phase G.

Proof. Coordinate unit moves are optimal in phase C by the triangle inequality. In phase G, switching immediately and completing gives $1 + \|x\|_1$. Any distinguished prefix has displacement $U \geq 0$ coordinatewise and cost $\|U\|_1$, so every path through such a prefix costs at least

$$\|U\|_1 + 1 + \|x + U\|_1 \geq 1 + \|x\|_1.$$

The a_i commute as translations. Their private p_i coordinates distinguish every subset of actions, proving structural nondegeneracy. □

For $J \subseteq [s]$ put $U_J = \sum_{i \in J} u_i$. The incidence identity is

$$(U_J)_{h_E} = |J \cap E|, \qquad (U_J)_{p_i} = \mathbf{1}_{i \in J}.$$

Since $v_A \leq 0$ and $U_J \geq 0$ coordinatewise, the distinguished face J is geodesic exactly when

$$U_J \leq -v_A$$

coordinatewise.

Theorem 5.2 (Universal k-fiber witness)**.** *The 2^m states v_A, $A \subseteq \mathcal{E}$, have one common exact distance and one common complete optimal-compatibility truncation through order $k-1$, while*

$$J \in H_{=k}(v_A) \quad \iff \quad J \in A$$

for every $J \in \mathcal{E}$. Hence every one of the $2^{\binom{s}{k}}$ order-k patterns on the distinguished s-simplex is realized inside one lower-order fiber, attaining the combinatorial maximum.

Proof. First,

$$\|v_A\|_1 = (m + 1 - |A|) + s + \sum_{E \in \mathcal{E}} (k - 1 + \mathbf{1}_{E \in A}) = s + 1 + mk,$$

so every witness has distance $s + 2 + mk$ by Theorem 5.1. If $|J| \leq k - 1$, then at every incidence coordinate

$$|J \cap E| \leq k - 1 \leq -(v_A)_{h_E},$$

and private coordinates are never overshot; hence every lower-order distinguished face is geodesic for every A. The switch singleton is also always geodesic, and structural isolation of phase C supplies no mixed lower-order faces.

Now take $|J| = k$. At its own incidence coordinate h_J one has $(U_J)_{h_J} = k$. The available threshold is k exactly when $J \in A$ and only $k - 1$ otherwise. Every other incidence coordinate h_E with $E \neq J$ receives at most $k - 1$. Thus J is geodesic exactly when $J \in A$. All 2^m patterns occur, attaining the trivial combinatorial upper bound. □

5.3 From universal freedom to a common collapse

Turn geodesic compatibility into an ordinary deterministic semantic system by adjoining a dead state $\perp$. For a distinguished action a_i define

$$U_{a_i}(v) = \begin{cases} T_i v, & c_i + d(T_i v) = d(v), \\ \perp, & \text{otherwise}, \end{cases} \qquad U_{a_i}(\perp) = \perp.$$

A structural action word remains live exactly while it is a shortest prefix.

Lemma 5.3 (Affine prefix retention)**.** *For every executable action word w from a fixed phase ϕ,*

$$T_w(z, \phi) = \big(z + G_w(\phi), \tau_w(\phi)\big).$$

Hence T_w is injective in z. In particular, every lower-order structural prefix T_J preserves all distinctions among the universal witness states v_A.

Proof. Composition of translations is translation by the sum of the phase-dependent displacement vectors encountered along the fixed phase word. Subtracting the common displacement recovers the initial lattice coordinate. □

Theorem 5.4 (Universal face-collapse retention). *Let a full universal periodic k-fiber on an s-simplex be witnessed by $\{v_A : A \subseteq \binom{[s]}{k}\}$. Fix $|J| = k-1$, $j \notin J$, and $E = J \cup \{j\}$. Then the sources*

$$x_A = T_J v_A, \qquad E \notin A,$$

are pairwise distinct and all satisfy

$$U_{a_j}(x_A) = \bot.$$

Therefore every full universal periodic k-fiber has a one-action collapse family of exact size

$$\boxed{N^{\text{face}}_{s,k} = 2^{\binom{s}{k}-1}},$$

and retained information

$$\boxed{I_{\text{ret}} = \binom{s}{k} - 1 \ \textit{bits.}}$$

Proof. Exactly half of the 2^m patterns omit E. By Theorem 5.3, applying T_J preserves their pairwise distinctness. Full universality makes every $(k-1)$-prefix J geodesic and makes the completed k-face E nongeodesic exactly on the branch $E \notin A$. Since J itself is geodesic,

$$c_J + d(T_J v_A) = d(v_A).$$

Thus geodesicity of the next action a_j at $T_J v_A$ is equivalent to geodesicity of the whole face E at v_A. On the branch $E \notin A$ the geodesic-prefix semantic transition therefore goes to $\bot$. □

Corollary 5.5 (Fixed-action star amplification). *In the explicit universal construction, fix $j \in [s]$ and take all $(k-1)$-sets $J \subseteq [s] \setminus \{j\}$. Their collapse-source families are disjoint, and the common final action a_j has a collapse family of size*

$$\boxed{N^{\star}_{s,k} = \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.}$$

Proof. The post-prefix private-coordinate pattern records J: coordinate p_i equals 0 exactly when $i \in J$ and remains -1 otherwise. Thus source families for distinct J are disjoint. Each family has the face-collapse size from Theorem 5.4. □

6 Universal information forces physical exposure

Combining Theorem 3.3 with Theorem 5.4 gives the main representation-independent bridge.

Theorem 6.1 (Universal Information-to-Physical Exposure Theorem). *For every $s \geq k \geq 2$, every full universal periodic k-fiber, every chosen face-collapse interface, and every unrestricted deterministic exact ε-pushdown realization,*

$$\boxed{D + X_h + q_{\text{src}} \sum_{r=0}^{h} g^{r}_{\text{src}} \geq 2^{\binom{s}{k}-1}}$$

for every $h \geq 0$.

In particular, the global machine-resource bound

$$D + X_h + |Q| \sum_{r=0}^{h} |\Gamma|^r \geq 2^{\binom{s}{k}-1}$$

holds independently of the chosen stack encoding and normalization history.

Corollary 6.2 (Zero-debt bit budget)**.** *Assume zero debt and $g_{\mathrm{src}} = g \geq 2$. If every source has exposure at most h, then*

$$q_{\mathrm{src}} \frac{g^{h+1} - 1}{g - 1} \geq 2^{\binom{s}{k}-1}.$$

Equivalently, the exact depth bound is

$$h \geq \left\lceil \log_g \left(1 + \frac{(g-1)2^{\binom{s}{k}-1}}{q_{\mathrm{src}}} \right) \right\rceil - 1.$$

In particular,

$$\log_2 q_{\mathrm{src}} + (h+1) \log_2 g > \binom{s}{k} - 1 + \log_2(g-1).$$

Corollary 6.3 (Universal exposure spectrum)**.** *In the zero-debt case and $g \geq 2$,*

$$e_{(r)} \geq \left\lceil \log_g \left(1 + \frac{(g-1)r}{q_{\mathrm{src}}} \right) \right\rceil - 1, \qquad 1 \leq r \leq 2^{\binom{s}{k}-1}.$$

For fixed q_{src} and fixed $g \geq 2$, every fixed positive quantile has exposure $\Omega(\binom{s}{k})$. In particular, for fixed k this is $\Omega(s^k)$. For $g = 1$, fixed positive quantiles are exponential in $\binom{s}{k}$.

Corollary 6.4 (Star-amplified pushdown bound)**.** *For the explicit universal witness and fixed final action a_j,*

$$D + X_h + q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^r \geq \binom{s-1}{k-1} 2^{\binom{s}{k}-1}.$$

6.1 The graph case

For $k = 2$, the universal fiber realizes every graph on s distinguished vertices while exact distance and all order-1 compatibility data remain fixed. A single edge query retains

$$2^{\binom{s}{2}-1}$$

sources before collapse. Hence fixed finite control and fixed nonunary source alphabet force exposure $\Omega(s^2)$ on every fixed positive quantile. In the star-amplified construction, one final vertex action simultaneously supports

$$(s-1)2^{\binom{s}{2}-1}$$

collapse sources across its incident edge contexts.

Example 6.5 (The smallest nontrivial graph fiber)**.** *Take $s = 3$ and $k = 2$. Then $m = 3$, so one common order-1 signature contains all eight graphs on the three distinguished vertices. Fix the edge $E = \{1, 2\}$, use prefix $J = \{1\}$, and test the final action a_2. Exactly four graphs omit E; after the prefix their four semantic states remain distinct, and all four collapse to $\perp$ on a_2. Thus $I_{\mathrm{ret}} = 2$ bits.*

If a zero-debt realization uses one source control state and a binary source alphabet, the rank theorem gives

$$(d_1, d_2, d_3, d_4) = (0, 1, 1, 2).$$

Hence at least one of the four sources must expose two initial work cells. The example is intentionally small: the general theorem replaces these four sources by $2^{\binom{s}{k}-1}$.

7 Strong converse and simultaneous-query hardness

7.1 Strong converse

For a universal face-collapse family put

$$m = \binom{s}{k}, \qquad N = 2^{m-1}, \qquad C_h^{\mathrm{src}} = q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r}.$$

Then Theorem 2.6 becomes

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq \left(1 - \frac{C_h^{\mathrm{src}}}{2^{m-1}}\right)_+.}$$

If

$$\log_2 C_h^{\mathrm{src}} \leq m - 1 - \lambda,$$

then

$$\boxed{\frac{D + X_h}{2^{m-1}} \geq 1 - 2^{-\lambda}.}$$

Thus a deficit of λ physical observation bits forces all but a $2^{-\lambda}$ fraction of the retained semantic sources to pay canonical debt or deeper exposure.

7.2 A direct-sum theorem across all k-face queries

For every $E \in \binom{[s]}{k}$ choose once and for all a decomposition

$$E = J_E \cup \{j_E\}, \qquad |J_E| = k - 1,$$

and use the face-collapse query obtained by applying J_E and then testing a_{j_E}. For a fiber pattern $A \subseteq \binom{[s]}{k}$, the query E is an *absent-face query* when $E \notin A$. Call an absent-face incidence (A, E) *h-hard* when its realization either has canonical debt or requires exposure $> h$.

Assume one common upper bound $V(h)$ on the depth-h source observation count for every face query. Put

$$\eta_h = \frac{V(h)}{2^{m-1}}.$$

Theorem 7.1 (Simultaneous-query direct sum)**.** *There exists a universal witness pattern A with at least*

$$\boxed{\frac{m}{2}(1 - \eta_h)_+}$$

h-hard absent-face queries. In particular, if $V(h) \leq 2^{m-1-\lambda}$, then one witness is hard on at least

$$\boxed{\frac{m}{2}(1 - 2^{-\lambda})}$$

absent-face queries.

Proof. For each face E, exactly 2^{m-1} fiber patterns omit E. By the filtered capacity theorem, at most $V(h)$ of those absent-branch incidences can be both canonical and h-confined. Hence the number of hard incidences for this face is at least $\left(2^{m-1} - V(h)\right)_+$. Summing over the m faces gives at least

$$m\left(2^{m-1} - V(h)\right)_+$$

hard incidences. Averaging over the 2^m witness patterns yields one A with at least

$$\frac{m}{2}(1 - \eta_h)_+$$

hard queries. $\square$

Theorem 7.2 (Typical-case direct sum)**.** *Let A be uniformly random among the 2^m universal fiber patterns. For every $0 < \rho < 1/2$ and $0 < \zeta < 1/2 - \rho$, at least a fraction*

$$\boxed{\left(1 - e^{-2\rho^2 m} - \frac{\eta_h}{2\zeta}\right)_+}$$

of all patterns have at least

$$\boxed{m\left(\frac{1}{2} - \rho - \zeta\right)}$$

h-hard absent-face queries.

Proof. The number $Z(A)$ of absent faces is distributed as $\mathrm{Bin}(m, 1/2)$. Hoeffding's inequality [12] gives

$$\Pr\left[Z(A) < m\left(\frac{1}{2} - \rho\right)\right] \le e^{-2\rho^2 m}.$$

Across all patterns and faces, the number of easy absent incidences is at most $mV(h)$. Therefore the expected number $Y(A)$ of easy absent queries of a random pattern is at most

$$\mathbb{E}Y(A) \le \frac{mV(h)}{2^m} = \frac{m\eta_h}{2}.$$

Markov's inequality yields

$$\Pr[Y(A) > \zeta m] \le \frac{\eta_h}{2\zeta}.$$

Outside the union of these exceptional events, the number of hard absent queries is at least

$$Z(A) - Y(A) \ge m\left(\frac{1}{2} - \rho - \zeta\right).$$

□

Corollary 7.3 (Asymptotically almost-everywhere hardness)**.** *If $s \to \infty$ and the depth-h physical capacity deficit $\lambda = \lambda(s)$ tends to infinity so that $\eta_h \le 2^{-\lambda}$, choose for example $\rho = m^{-1/4}$ and, for all sufficiently large s, $\zeta = 2^{-\lambda/2}$. Then $\rho, \zeta \to 0$ and*

$$e^{-2\rho^2 m} + \frac{2^{-\lambda}}{2\zeta} \to 0.$$

Consequently, asymptotically almost every universal witness pattern is h-hard on

$$\left(\frac{1}{2} - o(1)\right)\binom{s}{k}$$

of its absent-face queries.

8 Related resource notions and novelty boundary

The present resource should be distinguished from several established lines of work.

Pushdown height. Pushdown-height complexity measures the maximum current height of the store, usually as a function of input length and under weak, accepting, or strong computation measures [2]. Our exposure depth instead counts how many cells of the *initial source representation* must become visible before one fixed semantic collapse is canonically realized. Administrative growth above the source is free in the exposure metric, while deleting an old source prefix is costly even if the total current stack is monotonically decreasing. Space measures for stack and checking-stack automata likewise concern store size rather than depth of read access to a fixed initial representation [6].

Push complexity. Push complexity counts push operations used to accept an input [3]. Theorem 4.3 shows that source exposure can be arbitrarily large while the collapse computation performs no pushes at all. The two resources are therefore incomparable at the level relevant here.

Descriptional complexity. Classical and modern descriptional-complexity work studies tradeoffs among control states, stack alphabets, normal forms, simulations, and determinization [4, 5, 11]. The present theorem does not concern the cost of translating one automaton description into another. It fixes a semantic collapse interface and proves that every exact top-access pushdown representation must pay through source-local state/alphabet capacity, deeper exposure, or debt.

Reversible pushdown computation. Reversible pushdown automata impose backward determinism on the whole machine [1]. The present machines need not be reversible. We instead use a local noncoalescence fact forced by untouched stack suffixes: a many-to-one semantic transition can be realized irreversibly, but distinct hidden source tails cannot disappear into one canonical target without being exposed or leaving debt.

Interior-stack access models. Stack automata permit a read head to move through the interior of the stack [7], while deep pushdown automata permit operations below the top [8]. These are strictly different storage interfaces from the ordinary pushdown model used here. Our exposure parameter does not grant interior access; it measures how far a standard top-access realization is forced to uncover its original source stack.

Information erasure. The broad connection between logically many-to-one maps, reversibility, and physical information loss is classical [9, 10] and is not claimed here. No thermodynamic lower bound is asserted in this paper; "physical complexity" refers to resources of an exact automata realization. The contribution is the sharp retained-information/exposure tradeoff for deterministic top-access pushdown realizations, together with a universal higher-order semantic family that forces a collapse load of $\binom{s}{k} - 1$ bits after all lower-order geodesic data have been fixed.

To the best of our knowledge, the specific notion of source-stack exposure depth, its sharp rank-by-rank lower envelope, and the collapse-retained-information/debt strong converse do not have a direct counterpart in the neighboring pushdown resource measures above. We do not use an absolute priority claim; the distinction is structural and theorem-level.

9 Conclusion

The main principle established here is

$$\boxed{\begin{array}{c}\text{retained semantic information}\\ \Longrightarrow \\ \text{physical local capacity, deep exposure, or representation debt}\end{array}}$$

For generic finite collapse systems, the filtered theorem $D + X_h + V(h) \geq N$ isolates the information-theoretic mechanism. For unrestricted deterministic ε-pushdown realizations, top access turns this into the sharp universal source-local capacity envelope

$$q_{\mathrm{src}} \sum_{r=0}^{h} g_{\mathrm{src}}^{r},$$

and the resulting rank-by-rank exposure lower envelope is pointwise sharp.

Universal periodic k-fibers supply a canonical high-information test family. Their lower-order prefix maps are injective translations, so fixing one k-face to the collapse branch leaves exactly $2^{\binom{s}{k}-1}$ distinct semantic sources. Thus the higher-order freedom that survives all lower-order optimality information cannot be hidden for free in an exact pushdown realization. The strong converse and direct-sum theorems show that the effect is not confined to a single exceptional source or a single query: a physical capacity deficit forces a large fraction of sources, and eventually almost every universal witness across roughly half of all face queries—equivalently, almost all of its absent-face queries—to incur debt or deep exposure.

References

[1] M. Kutrib and A. Malcher, *Reversible pushdown automata*, Journal of Computer and System Sciences 78 (2012), 1814–1827. `https://doi.org/10.1016/j.jcss.2011.12.004`

[2] G. Pighizzini and L. Prigioniero, *Pushdown and one-counter automata: Constant and non-constant memory usage*, Information and Computation 306 (2025), 105329. `https://doi.org/10.1016/j.ic.2025.105329`

[3] G. Pighizzini, *Push complexity: Optimal bounds and decidability*, Theoretical Computer Science 1071 (2026), 115839. `https://doi.org/10.1016/j.tcs.2026.115839`

[4] T. Masopust, *A note on limited pushdown alphabets in stateless deterministic pushdown automata*, International Journal of Foundations of Computer Science 24(3) (2013), 319–328. `https://doi.org/10.1142/S0129054113500068`

[5] L. Kiss and B. Rovan, *Descriptional complexity of push down automata*, ITAT 2020, CEUR Workshop Proceedings 2718 (2020), 59–66.

[6] O. H. Ibarra, J. Jirásek, I. McQuillan, and L. Prigioniero, *Space complexity of stack automata models*, International Journal of Foundations of Computer Science 32(6) (2021), 801–823. `https://doi.org/10.1142/S0129054121420090`

[7] J. E. Hopcroft and J. D. Ullman, *Nonerasing stack automata*, Journal of Computer and System Sciences 1(2) (1967), 166–186. `https://doi.org/10.1016/S0022-0000(67)80013-8`

[8] A. Meduna, *Deep pushdown automata*, Acta Informatica 42(8–9) (2006), 541–552. `https://doi.org/10.1007/s00236-006-0005-0`

[9] R. Landauer, *Irreversibility and heat generation in the computing process*, IBM Journal of Research and Development 5 (1961), 183–191. `https://doi.org/10.1147/rd.53.0183`

[10] C. H. Bennett, *Logical reversibility of computation*, IBM Journal of Research and Development 17 (1973), 525–532. `https://doi.org/10.1147/rd.176.0525`

[11] O. Martynova, *Exact descriptional complexity of determinization of input-driven pushdown automata*, preprint, 2024. `https://arxiv.org/abs/2404.10516`

[12] W. Hoeffding, *Probability inequalities for sums of bounded random variables*, Journal of the American Statistical Association 58 (1963), 13–30.